CSE 816: Software Production Engineering Course

Final Project Guidelines and Evaluation Criteria

Total Marks: 25

- **Project Expectations:**
  The final project requires students to design and implement a complete DevOps framework to automate the Software Development Life Cycle (SDLC) using appropriate DevOps tools. The implementation is expected to include:

  - **Version Control**: Git and GitHub
  - **CI/CD Automation**: Jenkins, GitHub Hook Trigger for GITScm Polling, and Jenkins pipelines
  - **Containerization**: Docker and Docker Compose
  - **Configuration Management**: Ansible Playbooks
  - **Orchestration and Scaling**: Kubernetes (K8s)
  - **Monitoring and Logging**: ELK Stack (Elasticsearch, Logstash, and Kibana)

Alternatively, students may choose equivalent tools to achieve the same objectives if justified appropriately.

- **Evaluation Expectations:**
  Your project will be evaluated based on its ability to simulate real-world DevOps workflows, showcasing automation, modular design, and scalability. The following functionalities are mandatory:

  1. Incremental updates to the Git repository should trigger automated processes, including:
     - Jenkins fetching and building the updated code.
     - Running automated tests.
     - Pushing the generated Docker images to Docker Hub.
     - Deploying the Docker images to a target deployment system.
  2. Upon refreshing the application, the new changes should be visible seamlessly.

3. Application logs must feed into the ELK Stack, and the Kibana dashboard should visualize these logs, providing insights into application activities.

❖ **Security and Advanced Features Encouraged:**
We strongly encourage incorporating security practices and advanced features such as:

- **Secure Storage**: Use tools like Vault to securely store sensitive credentials (e.g., usernames and passwords).
- **Modular Design**: Implement modular code, such as roles in Ansible Playbooks.
- **High Availability and Scalability**: Use Horizontal Pod Autoscaling (HPA) in Kubernetes for dynamic scalability.
- **Live Patching**: Implement live patching to update the application without downtime.

❖ **Marks Distribution:**

1. **Working Project** (20 Marks):
   - Fully functional and deployable project: **20 Marks**
   - Partially functional due to last-minute issues (project demonstrates substantial completion): **15 Marks**
2. **Advanced Features** (3 Marks):
   - Usage of Vault, Roles in Ansible, and Kubernetes HPA: **3 Marks**
3. **Innovation** (2 Marks):
   - Creative or innovative solutions implemented in the project: **2 Marks**
4. **Domain-Specific Projects**:
   - In place of generic full-stack or web-based applications, the projects will focus on specific application domains such as MLOps, AIOps, DevSecOps, Networking (including NFV implementations), Big Data, Healthcare, and Finance, thereby aligning DevOps practices with real-world domain requirements.

By following these guidelines, students are expected to demonstrate their understanding of DevOps methodologies and their practical application in automating complex SDLC workflows.